Bulletin of Monetary Economics and Banking

Volume 26
Number 0 *16th BMEB Call for Paper Special Issue*

Article 3

2-28-2023

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Recommended Citation
Ghosh, Saurabh and Ranjan, Abhishek (2023) "A MACHINE LEARNING APPROACH TO GDP NOWCASTING:AN EMERGING MARKET EXPERIENCE," *Bulletin of Monetary Economics and Banking*: Vol. 26: No. 0, Article 3.
DOI: https://doi.org/10.59091/1410-8046.2055
Available at: https://bulletin.bmeb-bi.org/bmeb/vol26/iss0/3

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Bulletin of Monetary Economics and Banking, Special Issue 2023, pp. 33 - 54
p-ISSN: 1410 8046, e-ISSN: 2460 9196

A MACHINE LEARNING APPROACH TO GDP NOWCASTING: AN EMERGING MARKET EXPERIENCE

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ABSTRACT

The growth rate of real Gross Domestic Product (GDP), as measured by the National Statistical Office of India, is an important metric for monetary policy making. Because GDP is released with a significant lag, particularly for the emerging market economies, this article presents various methodologies for nowcasting and forecasting GDP, using both traditional time series and machine learning methods. Further, considering the importance of forward-looking information, our nowcasting model incorporates financial market data and an economic uncertainty index, in addition to high-frequency traditional macroeconomic indicators. Our findings suggest an improvement in the performance of nowcasting using a hybrid of machine learning and conventional time series methods.

Keywords: GDP; Nowcasting; Random forest; Neural network.
JEL Classifications: C45; C52; C53; E58.

Article history:
Received : September 06, 2022
Revised : December 22, 2022
Accepted : January 10, 2023
Available Online : February 28, 2023
https://doi.org/10.21098/bemp.v26i0.2455

All views and opinions expressed reflect those of the authors and not necessarily of the Reserve Bank of India. Authors are grateful to Dr. K.P. Prabheesh (discussant) and the participants of 16th BMEB Conference participants for his valuable comments.

I. INTRODUCTION

Policymakers are required to make decisions in real time with incomplete information. However, the majority of key statistics, including GDP, are released with a lag. These lags are significantly larger in Emerging Market Economies (EMEs) than in the United States and other developed nations. In India, GDP data is released with a lag of two months, which is even longer than other developing countries, such as China and Indonesia. The lags in the data release result in uncertainties not only for policy makers but also for banks and other financial market participants. Nowcasting, an effective tool for shorter horizon forecasting, aimed at predicting the present, the very near future, and the very recent past (Banbura *et al*., 2010), is becoming more popular because it can help reduce some of these uncertainties and therefore is widely used by central banks and other financial institutions.

Nowcasting models extract information about the target variable from data that is available at a higher frequency or is released earlier than the target variable. The GDP nowcasting literature include both factor models and nonfactor models. Simple time series models, such as moving average or univariate autoregressive AR(1) models have been employed to generate GDP nowcasts and forecasts. The bridge equation model combines qualitative judgements with 'bridge equations' (Baffigi *et al*., 2004). There is another stream of literature on nowcasting GDP using the Dynamic Factor Model (DFM) framework, proposed in the Giannone *et al*. (2008). The DFM framework attempts to estimate the common component in the variation in the regressors.

Nowcasting models for EMEs are difficult due to unavailability of the high frequency data to cover different sectors of the economy. Several attempts to forecast India's GDP growth have been made in the literature. For instance, Bhattacharya *et al*. (2011) used a bridge model to nowcast India's quarterly GDP growth, and Bragoli and Fosten (2018), Iyer and Gupta (2019), and Bhadury *et al*. (2021) use DFM to nowcast the growth of GDP in India. In this paper, we make two novel changes, which we believe will be an addition to the literature, as it will significantly improve nowcasting. First, rather than conventional techniques that depend on assumptions on functional forms and distribution, we have adapted ML techniques. The ML models are getting increasingly popular in predicting outcomes/nowcasting. For instance, Richardson *et al*.'s (2021) results indicate that a simple ML model in the bridge equation format has a better GDP nowcast performance for New Zealand as compared with the DFM-based nowcast model. Second, considering the futuristic properties, we also explore financial market data (in addition to real economic indicators) and economic uncertainty index in this nowcasting exercise. We explore the combination of three approaches together, i.e., applying ML in DFM settings with financial and economic uncertainty data in addition to other high-frequency real economic indicators. We use ML methods such as Random Forest and Prophet models along with the traditional time series ARIMA model to nowcast GDP. Our findings suggest an improvement in the performance of nowcasting using a hybrid of ML and conventional time series methods. While conventional econometric methods as well as ML methods have been extensively used to predict time series data such as GDP, we have not come across an application-based paper that uses a hybrid of both conventional and ML methods.

The nowcast algorithm in this paper can be classified into four steps, which include: 1) selecting a set of high frequency indicators from the set of all the indicators in a way that minimises data over-fitting without compromising the information content; 2) forecasting selected high frequency data using appropriate methodology, such as ARIMA, Random Forest, or Prophet models; 3) combining all high frequency indicators into a single dynamic factor; and 4) calculating the GDP based on a bridge equation that uses lagged GDP and a dynamic factor component, using the AR framework or neural network methodology. These four steps give us several forecasts for the testing period. Our findings show that the Prophet model, when used with the neural network methodology, predicts the best results for our sample period. Throughout this analysis, our endeavour has been to extract as much information as possible from the data series while avoiding data noise that results in model over-fitting.

The rest of the paper is organised as follows. Section II provides a brief literature survey of the literature. Section III summarises the data. Section IV explains the methodologies. Section V sums up the results and compares different model projections and Section VI concludes.

II. LITERATURE

In the recent years, nowcasting techniques have become indispensable tools that help in designing pre-emptive and calibrated macro-financial policies. However, long lags in data availability for EMEs poses challenges for monetary policy decision making, which is forward looking in nature. Therefore, for well-informed policy decision making, it is imperative to rely on High Frequency Indicators (HFIs). Among the available techniques, AR models and DFMs are most commonly used for such nowcasts. In a bridge equation model, forecasts of each indicator are based on AR models. Then, quarterly GDP is nowcasted using these series. The DFM model is an improvement over the bridge equation framework (or simple regression methods) previously used to predict GDP growth based on a few monthly macroeconomic variables. Following Giannone *et al*. (2008), DFM has been used to nowcast GDP growth in the advanced economies, e.g. the United States (Lahiri and Monokroussos, 2011), France (Barhoumi *et al*., 2011), Germany (Marcellino and Schumacher, 2010), the euro area as a whole (e.g., Angelini *et al*., 2011), Norway (Aastveit and Trovik, 2012), Ireland (D'Agostino *et al*., 2013), and New Zealand (Matheson, 2010). Banbura *et al*. (2013) provides an excellent survey of this literature. In addition to real GDP, nowcast for nominal GDP are also estimated. For instance, Barnett *et al*. (2016) and Tang *et al*. (2020) have nowcasted nominal GDP incorporating Divisia monetary aggregates in the model. Bhattacharya *et al*. (2011) concluded that survey data in India marginally improves the performance of bridge regressions, in contrast to the higher predictive power in the advanced economies.

DFM based growth nowcast has also been applied to EMEs more recently, such as the PRC (Giannone *et al*., 2014); Brazil (Bragoli *et al*., 2014); Indonesia (Luciani *et al*., 2015); BRIC countries and Mexico (Dahlhaus *et al*., 2015), and for a group of EMEs including Brazil, Indonesia, Mexico, South Africa, and Turkey (Cepni *et al*., 2019). Zhemkov (2021) uses the accuracy of forecast combination of several

advanced nowcasting using expenditure components of GDP of Russia. For India, in particular, notable nowcasting attempts using DFM are those by Bragoli and Fosten (2018), Iyer and Gupta (2019), and Bhadury *et al*. (2021). Bragoli and Fosten (2018) compare nowcasts of the first release of GDP to those of the final release to assess the differences in their probabilities. Furthermore, they use nominal and international series to proxy the missing employment and service sector variables in India and find that the predictability increases for the final estimates. The informal sector's contribution to the GDP is relatively large, and this contribution is difficult to quantify. To tackle this challenge, it is important to look for survey data (Bhattacharya *et al*., 2011) or other proxy series such as nominal and international series as in Bragoli and Fosten (2018), or rainfall deviation as in Iyer and Gupta (2019). Iyer and Gupta (2019) show that rainfall has a high predictive content for India as a large proportion of labour force is involved in agriculture and allied activities, whereas monthly indicators are used to track each of the components of the economy. Finally, Bhadury *et al*. (2021) use HFIs of economic activity to nowcast Indian GDP. In the model, the authors first create a Monthly Activity Index based on carefully selected 15 HFIs using a DFM and then using the Activity Index in bridge estimation to get the nowcast of the GDP.

In this paper, following Bragoli and Fosten (2018) and Bhadury *et al*. (2021), we track the components of the economy including financial variables and introduce various methodologies for computing dynamic factor components and nowcasting GDP. We attempt to address some of the challenges associated with the nowcast exercise for an EME, which include HFIs selection from many potential indicators, correcting Jagged Edges in the HFI, and bridging frequency mismatch between quarterly GDP data and monthly HFIs etc., using new techniques. We hope that the analysis presented in this paper will help address some of these shortcomings.

III. DATA

We use quarterly GDP data and a slew of high-frequency monthly indicators (source in the Appendix, Table A.I). While all these variables are published by different agencies, we use either Reserve Bank of India's Database for the Indian Economy or CEIC as data sources. Data from January 2003 to December 2019 is used for our study, as it is a stable period of economic activity and avoids the abnormalities of the COVID-pandemic shock. In India, GDP data is released with a lag of two months. While other high-frequency indicators are released before quarterly GDP, they have an asynchronous release calendar, i.e., they come with a lag of a week to a month.

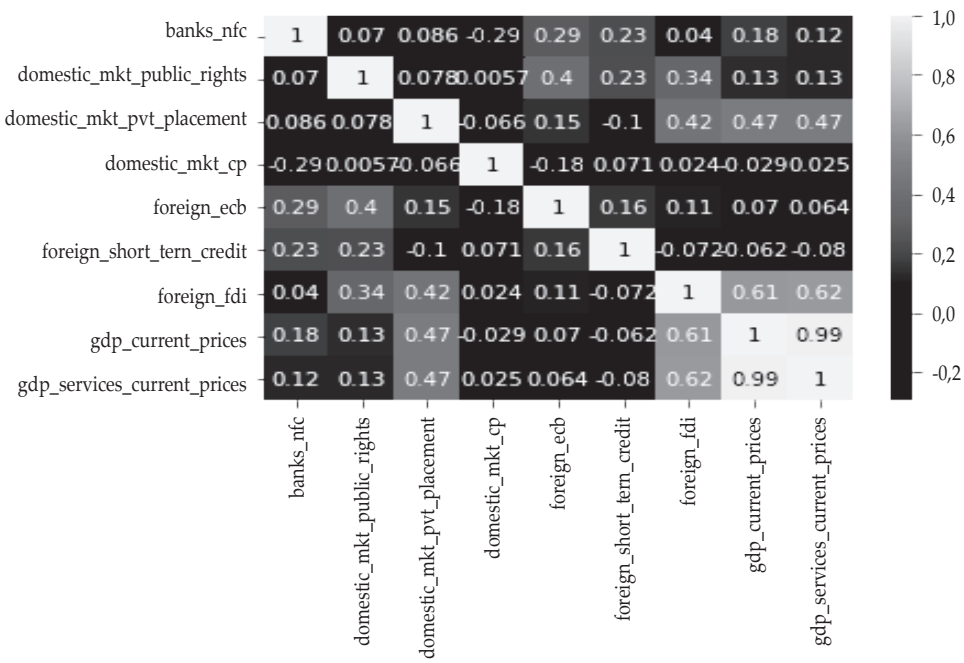
Our first task is to select a set of variables from the slew of high-frequency monthly indicators of economic activity. This selection assumes importance as large number of variables could lead to the "curse of dimensionality", ultimately resulting in over-fitting. The indicators are carefully analysed using the dynamic correlation, turning point analysis, and the Lasso method. These variables are same as that of Bhadury *et al*. (2021), as the sample time is same in their study. Further, we select financial variables and Economic Policy Uncertainty Index of Baker *et al*. (2015) to measure the effect of economic uncertainty to include forward looking information.

Financial time series are generally believed to quickly incorporate future information, even in a semi-efficient market setup (Fama and French, 1987). However, traditional GDP nowcasting models generally consider real variables rather than financial variables. In this section, we explore the possibility of some of the high-frequency variables that have a long time series and, if appropriate, may be used for nowcasting.

We use finances from domestic sources, such as bank non-food credit, rights issue, private placement of corporate debt, and domestic market capitalisation. Moreover, we also use finances from external sources, such as external commercial borrowing, short-term credit, and foreign direct investments. For consistency, we use the *Z*-score of all these series and the heatmap of the cross-correlation of these variables with financial sector GDP and GDP at current market prices is shown in Figure 1.

Figure 1. Pairwise Cross-Correlations

This figure reports pairwise cross-correlations between financial variables and FDP.



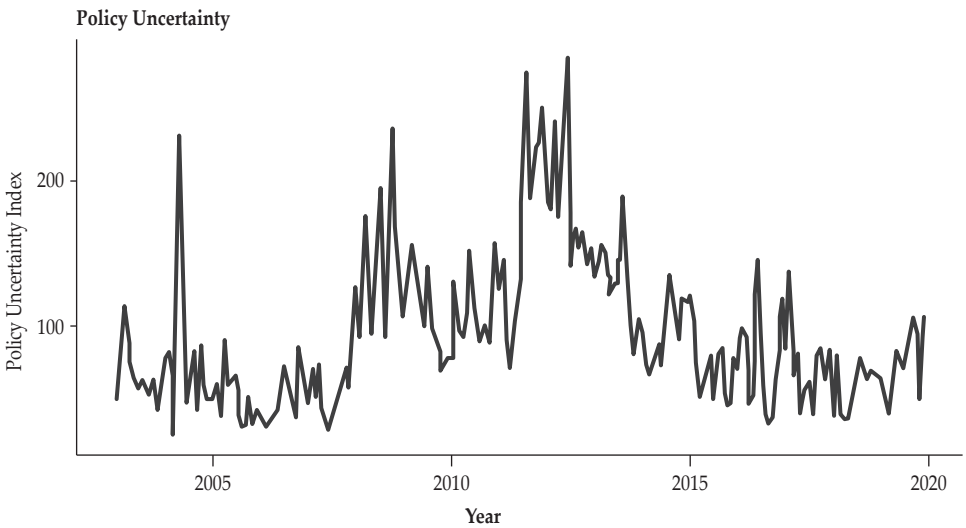
Furthermore, the pairwise distribution plot and the scattered diagram are reported in Figure 7. To avoid the *curse of dimensionality* and to pick up the most relevant financial variables, we use the Lasso technique to short-list a few of these financial variables. Our Lasso results show non-zero coefficients for two of our seven financial variables, namely domestic market private placement and foreign FDI flows. It may be mentioned that the variables, e.g., bank non-food credit, assume paramount importance in the Indian context. However, non-food credit may be having an impact on GDP with a lag and therefore is not highlighted by

the Lasso (and ridge) regression set-up. In this section, since our primary interest is to fine-tune GDP nowcasting, we mainly concentrate on the variables that have been shortlisted by the Lasso (and ridge) variable selection methodologies.

Finally, as a forward-looking gesture, we include the role of uncertainty in the forecasting model by using the Baker *et al*. (2015) uncertainty index (Figure 2). The authors used articles from seven newspapers to construct the policy uncertainty index. We use their monthly index as one of the variables in our Dynamic Factor-10 (DF10) and DF12 estimations.

Figure 2. Economic Policy Uncertainty Index

This figure shows Economic Policy Uncertainty Index for India (as reported in *Baker et al.* (2015)).



After selecting the variables, we use a three-step forecasting model in line with the DFM based nowcasting literature. The first step is to nowcast or forecast values for each economic indicator based on the historical values of those indicators. For, this we use three different models: (1) ARIMA Model; (2) Random Forest; and (3) Facebook's Prophet Model. Second, after predicting the values of the economic indicators, a single dynamic factor is extracted from the set of indicators. Finally, we use linear regression and neural network-based regression to calculate GDP. We will discuss each of these methods in the following subsections and present the results at the end of them.

Data on the financial variables as well as the uncertainty index are immediately available, and therefore, we do not need to nowcast these variables. We nowcast the remaining variables using the ARIMA, Random Forest, and Prophet models.

IV. METHODOLOGY

A. ARIMA Model

The ARIMA method is likely one of the most time-tested techniques for forecasting univariate time series data, which may not call for an elaborate description. We use

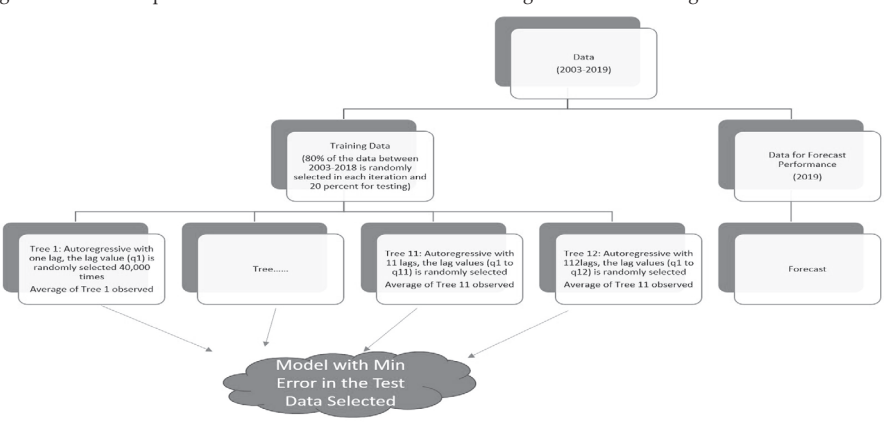
ARIMA forecasting as a baseline, by following the standard three-step procedure. Here, we employ the method described in Hyndman and Khandakar (2008). Each of the nine shortlisted data series is subjected to the Canova-Hansen test for seasonality, the KPSS-unit root test for integration, and if necessary, a difference operation on the series to render it stationary. The model's optimal lag structure is then determined using the appropriate information criterion (Akaike Information Criterion in our case). We then estimate the model parameters using the training sample and forecast the series for the test period using a dynamic forecasting algorithm. The actual and forecast are reported in Figure 4.

B. Random Forest

A random forest is a supervised ML algorithm that is constructed from decision tree algorithms. It builds decision trees on different samples and takes their majority vote for classification and average in the case of regression. Figure 3 shows the design of the Random Forest Regression for this study. Each tree predicts a value based on the tree algorithm, and then the average value is given as the final output or prediction.

Figure 3. Random Forest Algorithm

This figure shows the steps taken to forecast economic indicators using Random Forest Algorithm.



For each decision tree, past one year data or 12 lagged values are taken as predictors, and then decision trees are made using these lagged variables. As shown in Figure 3, 80 percent of the data for training is drawn from the period 2003-2018. At next sub-tree, a regression is run with $m \in \{1,2,\cdots,12\}$ number of randomly selected variables from the 12 lagged variables.

This step is repeated 40000 times for each m, and then the error is calculated on the average of these 40000 regressions. The m with the minimum error is selected for the prediction of the 2019 data. The resulting m can be the same or different for different variables/economic indicators. Each variable is predicted using an m number of randomly chosen lagged values from a pool of 12 lagged values, and the process is repeated 40000 times. The prediction value is taken as an average of these 40000 values.

C. Prophet Model

Prophet is a procedure for forecasting time series data based on an additive model where non-linear trends are fitted with yearly, weekly, and daily seasonality, plus holiday effects (Taylor and Letham, 2017). It works best with time series that have strong seasonal effects and several seasons of historical data.

This is a decomposable time series model described in Harvey and Peters (1990) with three main model components: trends, seasonality, and holidays. They are combined in the following equation:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{1}$$

where $g(t)$ represents non-periodic changes in the value of the time series, $s(t)$ represents periodic changes (e.g., weekly and yearly seasonality), and $h(t)$ represents the effects of holidays occurring on potentially irregular schedules over one or more days. The error term ϵ_t represents any idiosyncratic changes which are not accommodated by the model.

This specification is similar to a generalized additive model (Hastie and Tibshirani, 1987), a class of regression models with potentially non-linear smoothers applied to the regressors. In this model, only time is used as a regressor, but possibly several linear and non-linear functions of time as components. Modelling seasonality as an additive component is the same approach taken by exponential smoothing (Gardner Jr., 1985). Multiplicative seasonality, where the seasonal effect is a factor that multiplies *g(t)*, can be accomplished through log transformation.

In this model, unlike ARIMA, the measurements do not need to be regularly spaced, and we do not need to interpolate missing values, e.g., from removing outliers.

D. Comparing the Forecast of Economic Indicators

For our forecasting exercise, 2003–2018 data is taken as our training set, and 2019 is our test set. Figure 3 compares the actual 2019 data to the predicted value using three methods: ARIMA, Random Forest, and Prophet Model. As mentioned, 2003–2018 data was taken for developing the models, and the forecast for 2019 is calculated based on these variables. We give an example of Auto Sales for illustration—all other variables are predicted similarly.

The auto data is from 2003 to 2018. We used the *auto.arima* function from the forecast package (R) to calculate the best-fitting ARIMA model and then predict the values for the next 12 months. For Random Forest, we use the *randomForest* function from the *randomForest* package (R), and then predict the next 12 months. For the Prophet model, we use the *Prophet* package to develop the model, and then the *forecast* package to predict the forecast for the next 12 months.

We compared the forecasted value against the observed value for these variables. Based on the data in Table 1, we find that the ARIMA and Prophet models often outperform Random Forest (except IIP Consumer Goods), whereas ARIMA and Prophet are often close. In Table 1, the smaller values indicate the best performer, while those with bigger values are the worst performers.

Table 1.
Results Table

This table presents the mean square error for different economic indicators using ARIMA, Random Forest and Prophet Algorithm.

Indicator	MSE		
	ARIMA	RF	Prophet
Air Cargo	1823	2242	1724
Auto Sales	99600	114843	102344
Government Receipts	95810	112939	113198
IIP Cons	2.35	1.85	2.03
NONG Import	546	588	419
Rail Freight	1.53	2.14	1.46
Foreign Tourist	12775	19204	24440
IIP Core	1.32	1.65	1.59
Foreign FDI	11571	10901	9548

Figure 4. Comparison of Forecasts

This figure compares forecasts of the economic indicators using ARIMA, Random Forest and Prophet Model.

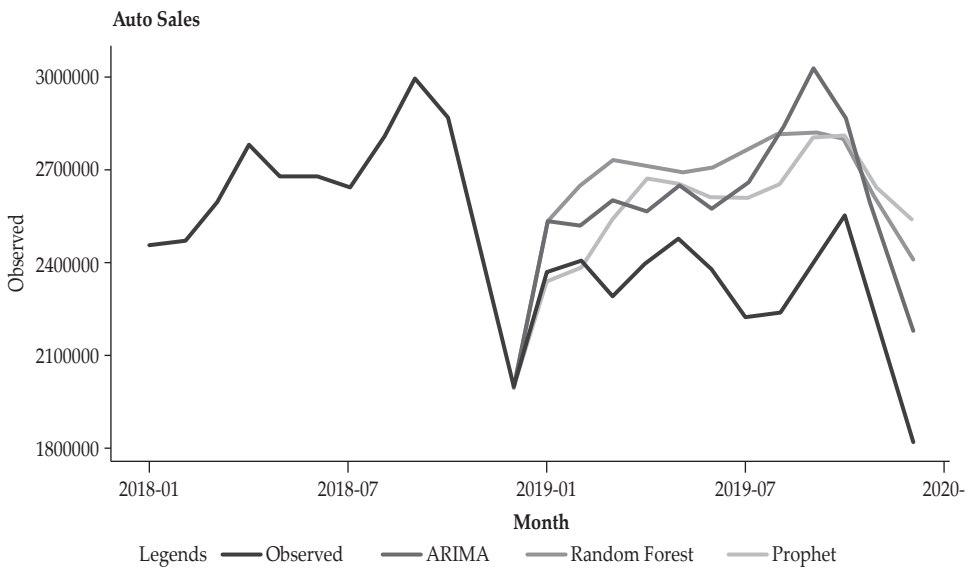
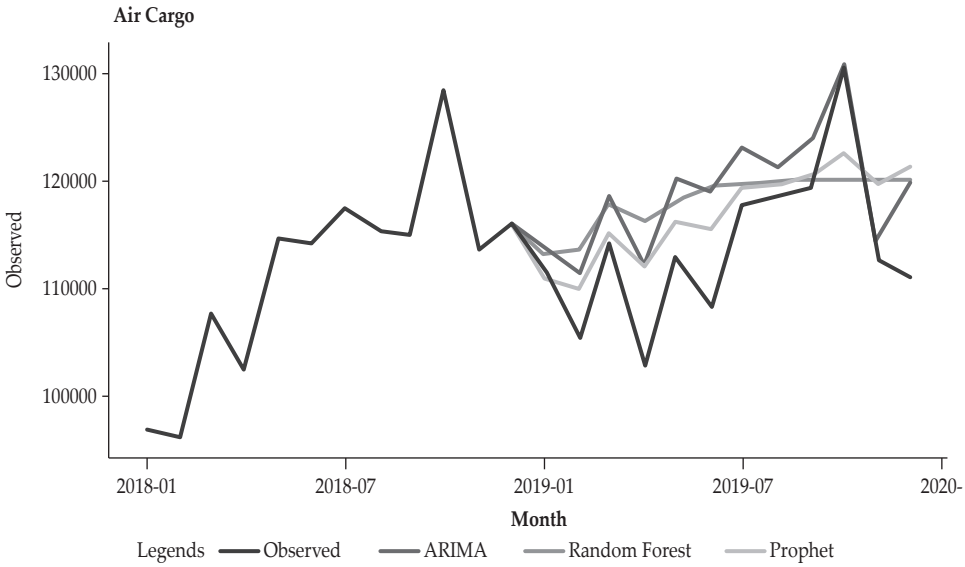


Figure 4. Comparison of Forecasts (Continued)



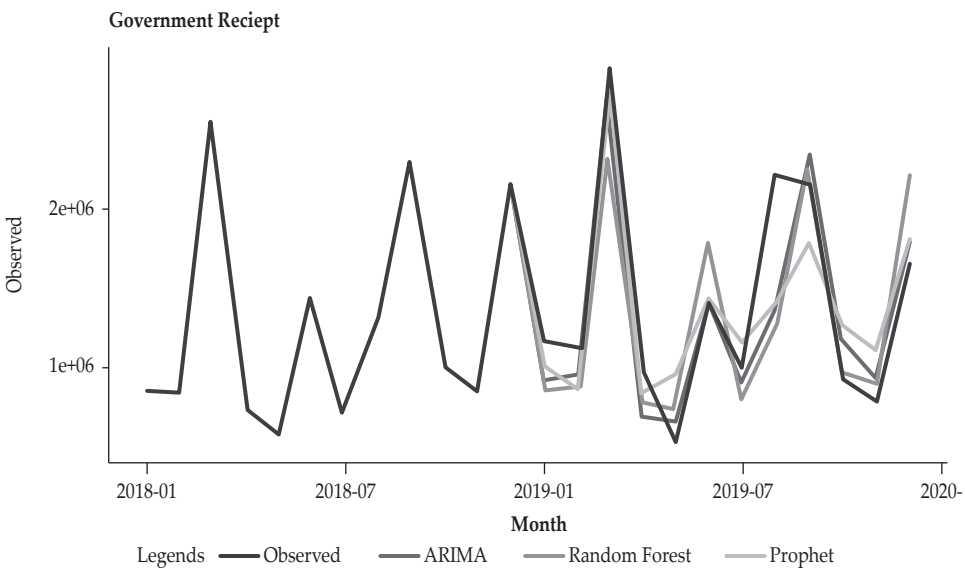
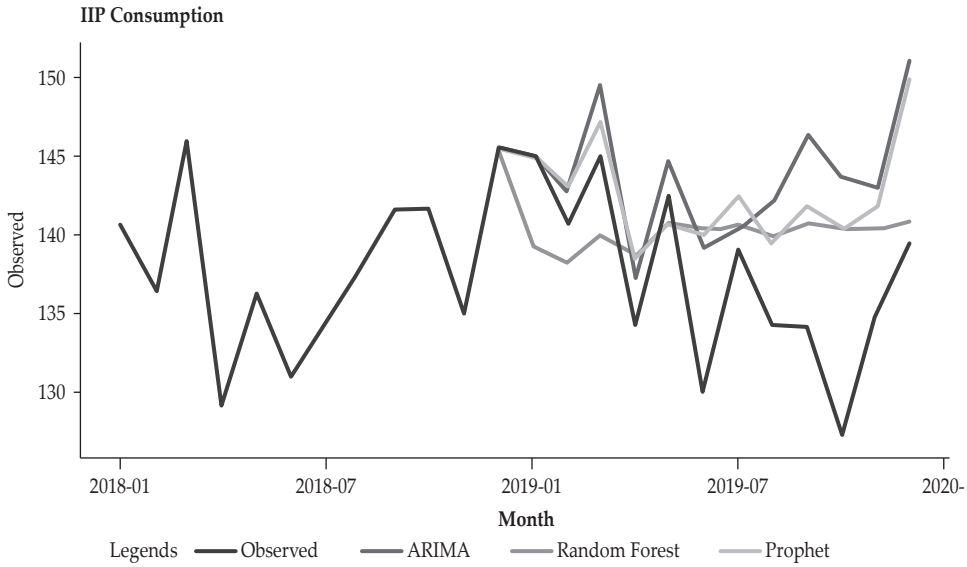


Figure 4. Comparison of Forecasts (Continued)



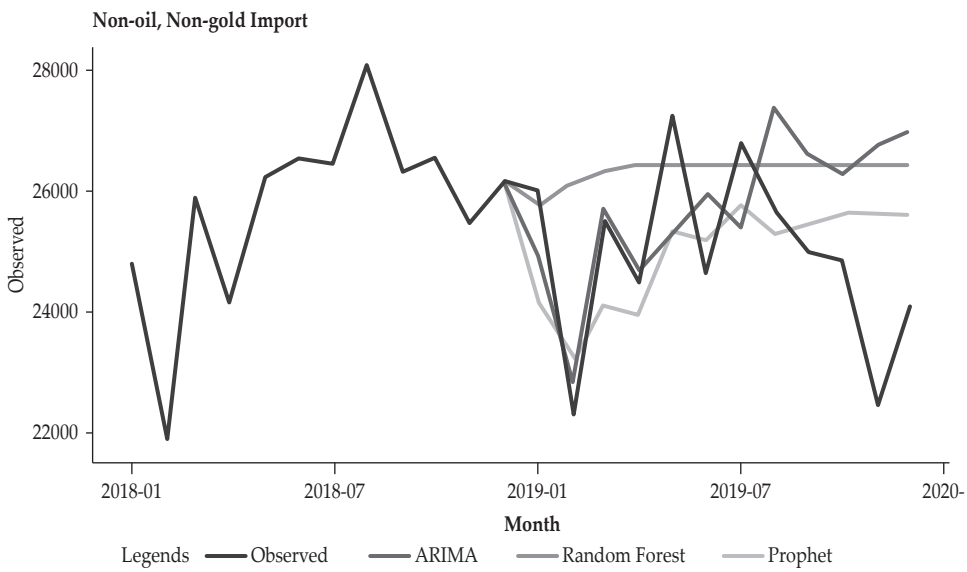
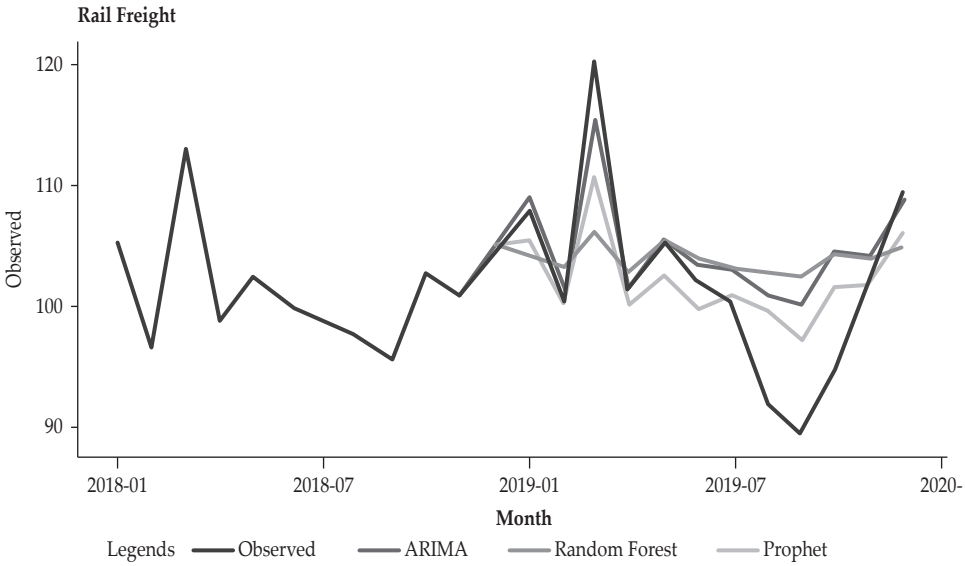


Figure 4. Comparison of Forecasts (Continued)



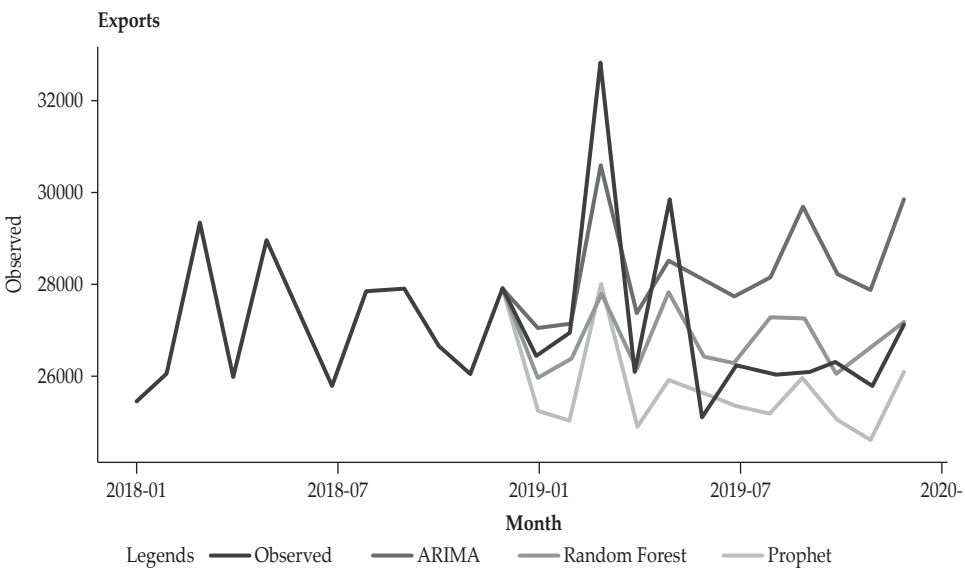
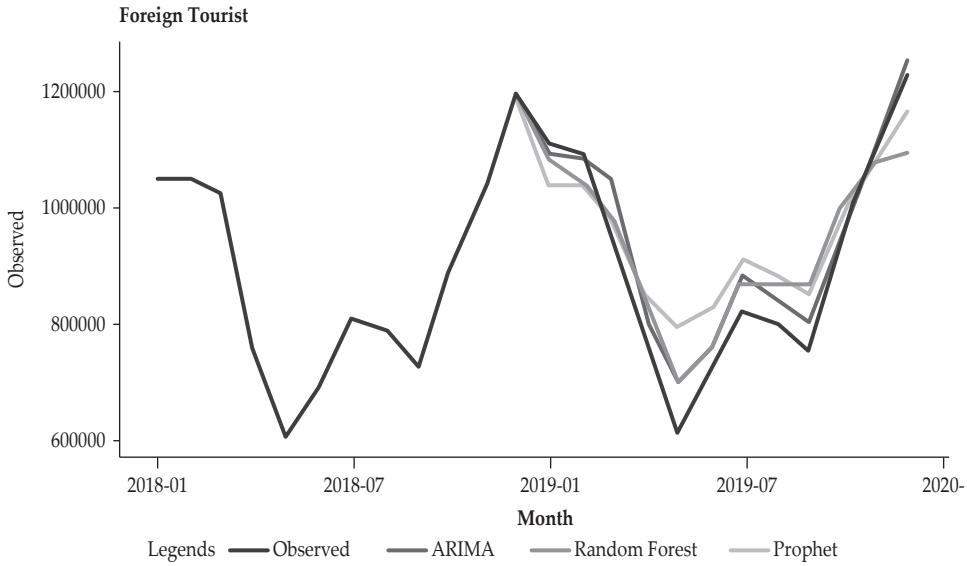
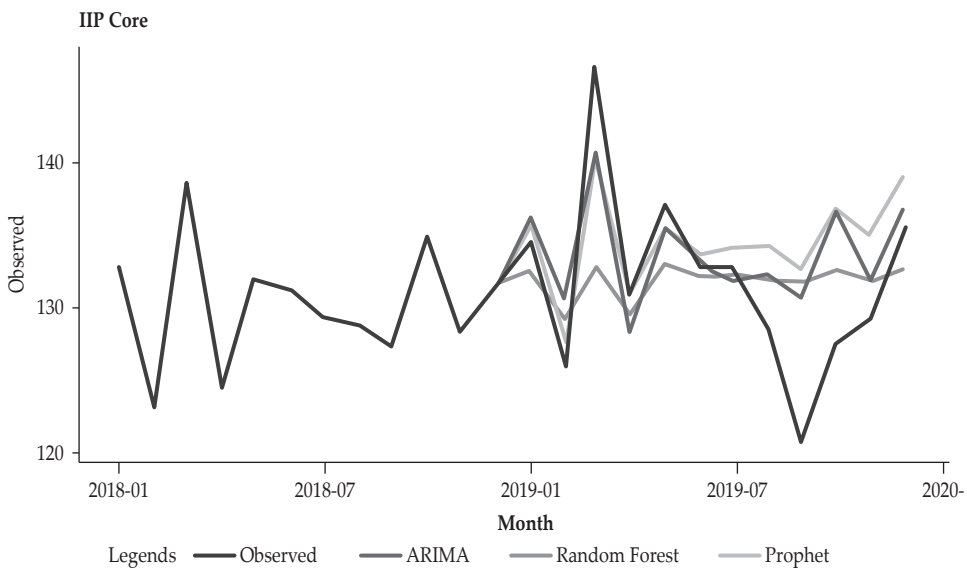


Figure 4. Comparison of Forecasts (Continued)





E. Dynamic Factor(s)

The DFA is a multivariate time-series analysis that allows the estimation of underlying common trends in short and nonstationary time-series. It has been frequently applied in GDP literature—see, e.g., Stock and Watson (1989), Giannone *et al.*, (2008), and Bhadury *et al.*, (2021). The DFA technique resembles factor analysis in the sense that new variables (factors) are created that, although they may be fewer in number than the original variables (one in our paper), are expected to

explain most of the variation. The main feature of DFA is that the new axes (CTs in time-series) are smooth functions (curves) over time, with smoothing estimated automatically as part of a two-step Expectation–Maximization (EM) algorithm. The conditional expectation of the log-likelihood function is calculated in the first step, then the function is maximised over the entire set of iterations (see details in Zuur *et al*., 2003). We use the estimated output obtained from the ARIMA, Random Forest, and Prophet model estimations to estimate dynamic factors; the factors are plotted in Figure 5. We log-transform the data and then calculate the *z*-score before applying the DFA technique. This helps to avoid any large-scale effect of the data.

Figure 5. Comparison of Forecasts

This figure compares dynamic factors computed using 6 and 9 economic indicators respectively for ARIMA, Random Forest and Prophet forecasts.

F. Auto-Regressive Model

In this section, we connect all of the above building blocks to nowcast GDP. For this, we need a bridge equation to connect our estimated dynamic factor to GDP. Our bridge equation is an AR(1) representation of GDP augmented by the DF estimates, which is as follows:

$$GDP_t = \alpha + \beta * GDP_{t-1} + \delta * DF_t + \epsilon_t \tag{2}$$

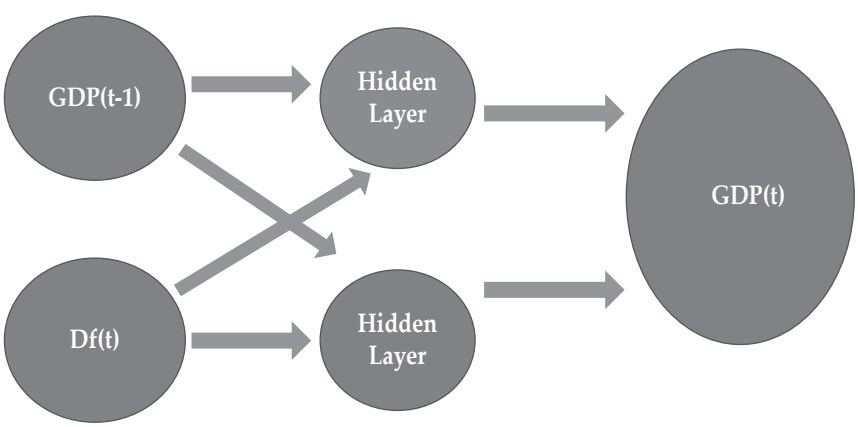
G. Artificial Neural Network

As an alternative to the above approach, we can also forecast GDP using artificial neural network methodology. Figure 6 shows how neural network models operate. In this model, we predict GDP using the dynamic factor components, and the lag of GDP as the predictors. To calculate GDP, we divide our data into training and testing sets. A training set is used to train the neural network model, and a testing set is used to evaluate the error. This method is also known as the *out of sample error test* to measure and reduce overfitting.

A neural network is a network consisting of several hidden layers, and at each layer, there can be several nodes. Each of these nodes is computed using the nodes from the previous hidden layers. A final output is computed using the nodes in the last hidden layers. Usually, these nodes cannot be interpreted. It uses the *back propagation* method to compute the weights at each node. We use the *hit-and-trial* method to compute the number of hidden layers and nodes at each layer. For our data, the model estimates suggest one hidden layer with two neurons at the hidden layer. Finally, the model with the least error on the testing set is used for nowcasting GDP. Tables 2-5 show the GDP nowcasts that were generated using these methods.

Figure 6. Neural Network Algorithm

This figure shoes the steps of neural network with hidden levels.



V. RESULTS

In this section, we discuss the results obtained from combining DFMs with different methodologies. Our primary objective is to identify a set of algorithms that would identify the best nowcast and forecast. Following the common practice in this strand of literature, we use our test data (2019) to evaluate the nowcast and forecast.

As mentioned before, GDP growth in India is released with a lag of two months, whereas policy makers need to access the economy on a real-time basis and update their beliefs at regular intervals (with the release of every data point). The data is released with different lags (Table 6). To help policy makers update their beliefs, we access our nowcast model based on estimates obtained after seven days and 30 days from the end of the quarter. The training set for the nowcast is 2003–2018, and the test set is 2019. The out of sample rolling Root Mean Square Error (RMSE) is calculated for each model, and the results are summarised in Tables 2 and 3. The RMSE is calculated using the formula $\sqrt{(\frac{1}{4}\sum_{i=1}^{4}(x_i - \hat{x}_i)\text{^}2)}$. In Figure 7, we show the plot of DF12 based GDP nowcast using different nowcasting methods against the actual GDP. The left panel shows the comparison when the bridge equation is DF augmented AR (1) process, whereas the plot in the right panel shows the comparison when the bridge equation is the ANN model.

Figure 7. GDP and DF12 based GDP Nowcast

This figure shows the comparison of GDP nowcasts using Autoregressive and Neural Network models with observed GDP.

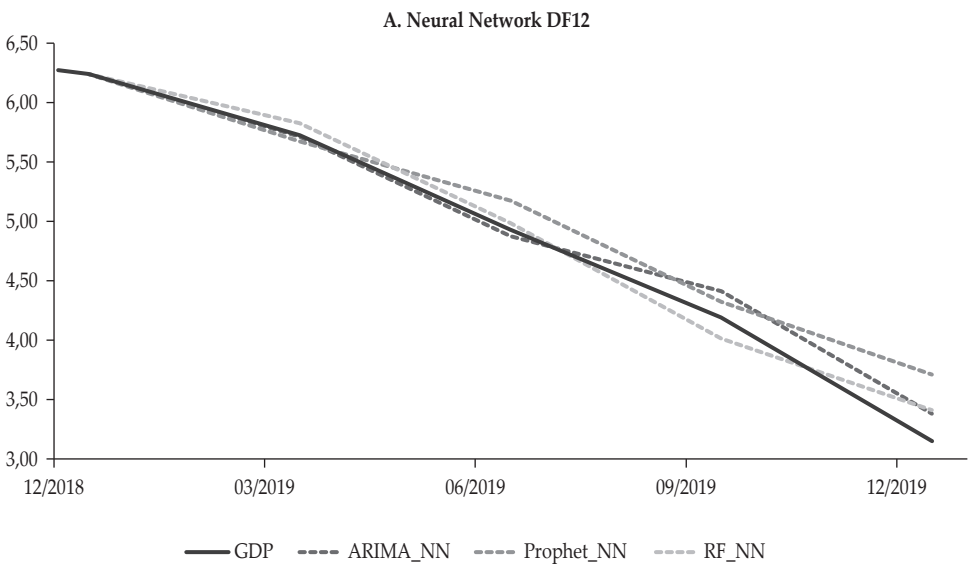


Figure 7. GDP and DF12 based GDP Nowcast (Continued)

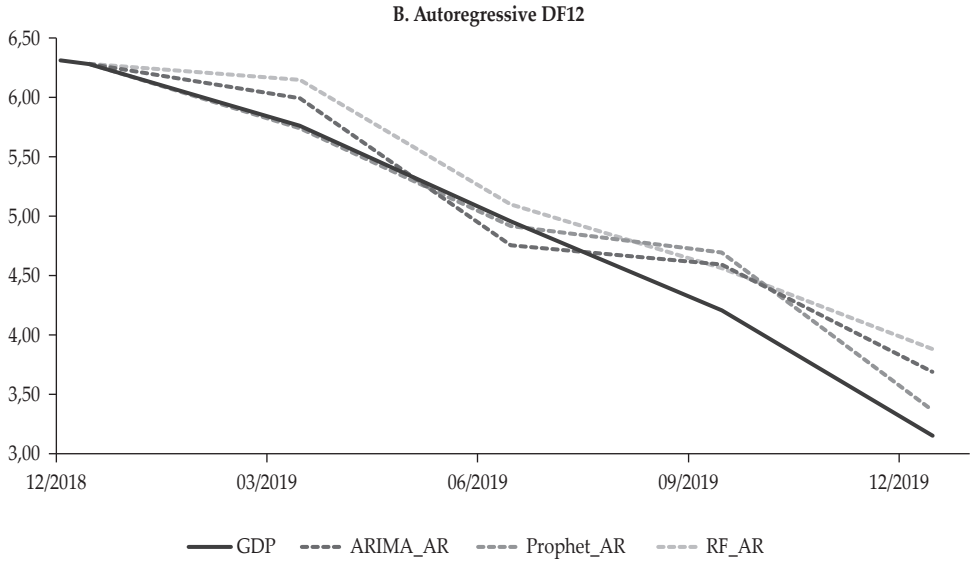


Table 2 (Rolling RMSE seven days after) shows the Prophet model often outperforms other models. It also shows that the information from the forward-looking variables contributes to significant improvement in the nowcast. Table 3 (Rolling RMSE 30 days after) shows a significant improvement when a neural network is applied to the bridge model instead of the AR model. Tables 2 and 3 both indicate a slightly better result using the Prophet model.

We further look at the forecast at one-year ahead horizon. We find that the RMSE for GDP forecast increases significantly as shown in the Table 4. Table 5 shows the RMSE for the two-quarters ahead horizon. We find that the error increases as we increase the forecasting horizon at any time, implying that these models have significant advantages, while nowcasting the GDP and may not be suitable for forecasting GDP.

Table 2.
Rolling RMSE Seven Days After

This table provides root mean square error for the computation of GDP using different algorithm when computed 7 days after the end of the quarter.

Model	Random Forest		ARIMA		Prophet	
	AR	NN	AR	NN	AR	NN
DF6	0.42	1.81	0.37	1.79	0.27	0.27
DF9	0.51	0.32	0.41	0.49	0.20	0.72
DF10	0.50	0.20	0.41	0.33	0.20	0.26
DF11	0.45	0.23	0.37	0.31	0.26	0.32
DF12	0.45	0.16	0.36	0.16	0.26	0.31

DF6: Air Cargo, Auto Sales, Government Receipts, IIP Cons, NONG Import, Rail Freight
DF9: DF6 + Exports, Foreign Tourist Arrival, IIP Core
DF10: DF9 + Economic Policy Uncertainty
DF11: DF9 + Domestic Market Private Placement, Foreign FDI Flows
DF12: DF11 + Economic Policy Uncertainty

Table 3.
Rolling RMSE One Month After

This table provides root mean square error for the computation of GDP using different algorithm when computed 7 days after the end of the quarter.

Model	Random Forest		ARIMA		Prophet	
	AR	NN	AR	NN	AR	NN
DF6	0.38	0.75	0.20	0.10	0.14	0.27
DF9	0.62	0.44	0.45	0.20	0.41	0.25
DF10	0.60	0.23	0.44	0.22	0.39	0.21
DF11	0.36	0.19	0.30	0.20	0.26	0.20
DF12	0.36	0.19	0.29	0.18	0.25	0.26

Table 4.
RMSE (Four Quarters)

This table provides root mean square error for the computation of GDP using different algorithm when computed 4 quarters ahead.

Model	Random Forest		ARIMA		Prophet	
	AR	NN	AR	NN	AR	NN
DF6	1.15	0.77	1.43	1.10	0.87	0.85
DF9	0.91	0.72	1.23	0.98	0.95	0.72
DF10	0.92	0.48	1.24	1.29	0.95	1.05
DF11	0.62	1.00	1.33	1.37	0.99	1.04
DF12	1.01	1.06	1.34	1.38	0.99	0.69

Table 5.
RMSE (Two Quarters)

This table provides root mean square error for the computation of GDP using different algorithms when computed 2 quarters ahead.

Model	Random Forest		ARIMA		Prophet	
	AR	NN	AR	NN	AR	NN
DF6	0.46	0.33	0.69	0.41	0.27	0.24
DF9	0.34	0.33	0.51	0.30	0.34	0.21
DF10	0.35	0.63	0.53	0.55	0.34	0.39
DF11	0.48	0.37	0.59	0.60	0.37	0.38
DF12	0.38	0.38	0.60	0.61	0.37	0.21

VI. CONCLUSION

Assessing the state of the economy in real time is a prerequisite for implementing suitable policy measures. The efficacy of policymaking is contingent upon the utilisation of all available information at any given time, and including forward looking information, such as stock market performance and economic policy uncertainty, available at the point of estimation. With changing economic phases and the availability of new data and techniques, the traditional time-series-based GDP forecasting had to be reconsidered. Our paper assesses the impact of these changes in a traditional nowcasting framework.

In this article, we estimate a variety of alternative nowcasts and assess their out-of-sample forecasting performance. In doing so, we combine conventional time series models with ML techniques (e.g., prophet forecasting, artificial neural networks, etc.). In addition, we utilise forward-looking variables (such as financial indicators and an uncertainty index based on newspaper articles) that have not been used in the conventional GDP nowcasting framework. Our findings indicate that the hybrid model provides the most accurate GDP forecasts for our sample period.

Any improvement in the nowcast is regarded both practical and desirable in a dynamic context. In addition, utilising the current flow of information is vital to the formulation of a forward-looking policy. Due to delays in the official data release of major macroeconomic variables, such as GDP, this is even more crucial. We believe our findings have the potential to improve macroeconomic variable nowcasts, facilitate policy decisions, and stimulate future research.

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APPENDIX

A.I. Pairwise Distribution and Scatter Plot

Figure A.I. Pairwise Plot Financial Variables and GDP

This figure shows the pairwise scatter plot for the financial variables.

A.II. High Frequency Indicators

Table A.I.
Data Source

This table provides sources of the data used in the paper.

Indicator	Source	Release Lag
Air Cargo	CEIC	3-4 weeks
Auto Sales	CEIC	12 days
Government Receipts	CEIC	1 month
IIP Cons	CEIC	1-month and12 days
NONG Import	CEIC	15 days
Rail Freight	CEIC	9-10 days
Exports	CEIC	1-2 days
Foreign Tourists	CEIC	17 days
IIP Core	CEIC	1 month
Domestic Market Private Placement	PRIME DATABASE	1 day
Foreign FDI	CEIC	1 month and 3 weeks
Baker's Uncertainty Index	Baker *et al.* (2015)	1 day